# Unchosen Routes Before Report: A Perturbational Test of Admissible Report-Route Breadth in Delayed Visibility Judgments

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## Author response and substantive revision

The two independent reviews identified a decisive defect in the original formalism. The former statistic summed lag-specific output outer products generated by a single content input and interpreted the effective rank of that sum as the number of independently reachable report policies. That interpretation was not warranted. Two selectively perturbable routes with proportional temporal profiles could produce rank one, while a single route whose measured loading rotated across output coordinates could produce rank greater than one. Lag-specific whitening also invalidated the claimed general output-coordinate and duplication invariances.

The former lag-summed “content-to-policy output-reachability Gramian” is therefore withdrawn. It is not retained as a primary mediator, a causal-rank measure, or an awareness criterion. The present revision replaces it with an **interventionally validated route-participation breadth** statistic. The replacement has four defining features:

1. candidate routes are established in independent, outcome-blind perturbation localizers rather than discovered from the rank of a one-dimensional content mapping;
2. route trajectories are embedded in one fixed output space over one fixed pre-final-cue horizon;
3. a single, treatment-independent, full spatiotemporal noise covariance is used rather than separate lag-specific whitening matrices; and
4. the resulting spectral statistic measures the balance and noise-adjusted participation of already validated routes. It does not establish causal independence by itself and is not called causal rank.

The reviews also correctly observed that randomizing physical orientation does not amount to intervening on a latent decoded neural content coordinate. The revision consequently distinguishes the randomized physical stimulus $U$, latent sensory evidence $E$, decoded neural content $Z$, and route-specific neural output coordinates. Average stimulus-linked route responses can receive a causal interpretation under randomization and explicit modeling assumptions. Transfer from decoded neural content remains predictive unless stronger intervention semantics are available.

The primary causal estimand has likewise been narrowed. It is now the randomized active-stimulation-by-route-admissibility effect on a later visibility report when the stimulated route is not ultimately used. A positive result would show that perturbing a task-active but subsequently unselected response system changes later reporting in a task-state-dependent manner. It would not, by itself, demonstrate a constitutive gate for awareness, phenomenal consciousness, or an interventionally identified neural mediator. Neural breadth, total transfer, sensory evidence, task-set maintenance, memory, criteria, and postdecision processes are analyzed as separable candidate explanations.

One limited interpretive disagreement is retained. A task-state-dependent effect on an unselected route would not be equivalent to an ordinary effect on motor execution, because the route is never executed on the critical trials and the final content response is held fixed. It would therefore discriminate against the simplest actual-route-only account. The reviewers are nevertheless correct that such a result would remain compatible with prospective task-set maintenance, response-rule competition, verbal or spatial recoding, memory, confidence formation, and report formatting. The manuscript treats those alternatives as unresolved unless additional evidence distinguishes them.

Finally, the stored source records do not validate trial-level source-resolved route-breadth estimation, selective online stimulation of the proposed targets, artifact recovery during concurrent electrophysiology, or the required factorial trial counts. The closest control-theoretic antecedent is a preprint record reporting synthetic benchmarks and a preliminary ECoG application in four macaques under ketamine, not validation of the present human MEG–TMS estimator.[17] The intracranial prefrontal-stimulation review does not validate the proposed TMS targets or timing.[35] The program is therefore explicitly stage-gated. A confirmatory human test must not begin until simulations and treatment-blind pilots meet preregistered recovery, reliability, artifact, target-engagement, trial-yield, and operating-characteristic criteria.

# Abstract

Predictive processing can provide a systematic basis for relating neural organization to features of conscious perception while not itself constituting a complete theory of consciousness.[1] Active interoceptive inference further emphasizes embodied regulation, uncertainty, precision, emotion, and self-related inference, but the supplied record does not derive a criterion for report-policy selection.[2] Global-workspace and thoughtseed proposals describe competition, selection, and broad cognitive influence,[@src-paper-amara-kestrel-7-arxiv-10] while a recent control-theoretic preprint characterizes workspace mediation through reachability, observability, input-output alignment, effective dimensionality, and routed breadth.[17] These approaches motivate, but do not establish, the narrower question tested here: does the state of a currently admissible but ultimately unused report route affect later visibility reporting?

Participants would discriminate a near-threshold visual orientation. An exclusion cue would leave two report routes possible, and a later cue would select the final route. On critical trials, the final content report would always be manual, while a speech or saccadic route would either remain admissible during a preselection interval or already be excluded. Individualized stimulation would target the speech-associated or oculomotor-associated system. The primary causal estimand is the difference between active and sham stimulation when the target route is admissible minus the corresponding difference after it has been excluded.

The revised neural statistic is **interventionally validated route-participation breadth**, not causal rank. Candidate manual, speech, and saccadic routes must first be distinguished by selective perturbation in independent localizers. Their content-linked trajectories are then represented in a fixed route-by-time output space and evaluated with one full, treatment-independent spatiotemporal covariance. The entropy effective rank of the resulting route-participation matrix quantifies how evenly reliable stimulus-linked response is distributed over the validated admissible routes. Independent routes with identical temporal profiles remain distinct; temporal variation within one validated route cannot create extra routes; exact measurement duplication is invariant; and redundant cue mappings are quotient-merged before breadth is computed. These properties depend on the validated route partition and do not allow observational breadth to discover causal independence.

The experiment separates nested claims. First, route-specific stimulation may alter later visibility reports only while the unselected route remains admissible. Second, stimulation may selectively reduce target-route participation and route breadth. Third, breadth may predict reports beyond total transfer, sensory evidence, scalar activation, task-set decoding, and report criteria. Fourth, a trace-matched perturbation may distinguish spectral balance from total transfer. Only the first claim is directly identified by the principal randomization. Mediation and threshold claims remain assumption-dependent.

A positive result would support a role for prospective report-route state in delayed visibility reporting, not a measure of phenomenal consciousness. Sensitive null effects after independently demonstrated and condition-matched target engagement would falsify the task-defined unchosen-route hypothesis. Failure of estimator recovery, selective perturbation, trace matching, artifact control, or adequate trial yield would prevent the proposed mechanistic test from proceeding rather than being reclassified as evidence.

**Keywords:** reported visibility; flexible reporting; route admissibility; counterfactual task state; route-participation breadth; perturbation; TMS; MEG; confidence; system identification

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# 1. Research question, claims, and scope

## 1.1 Primary research question

The primary question is:

> **When a participant will ultimately report a percept manually, does perturbing a different report route change the later visibility report only if that route was still a valid future option at the time of perturbation?**

The comparison separates three trial properties that are often correlated:

1. **sensory evidence**, meaning information about stimulus orientation available to perceptual and mnemonic systems;
2. **actual-route state**, meaning the efficacy and preparation of the manual route eventually used on the critical trial; and
3. **prospective route state**, meaning the readiness and content linkage of another route that is valid before the final cue but is not ultimately selected.

The experiment does not assume that the third property is awareness. It asks whether that property has a randomized, task-state-dependent effect on later reports.

## 1.2 Nested hypotheses

The revised proposal separates claims that the original manuscript treated too closely.

### H1: Unchosen-route contribution

Active stimulation of a validated speech-associated or oculomotor-associated route will change later visibility reports more when that route remains admissible than after it has been excluded, even though the final content response is manual in both conditions.

This is the principal causal hypothesis.

### H2: Route-specific neural engagement

The same stimulation will selectively reduce stimulus-linked participation of the target route, with limited changes in the manual route, early sensory decoding, and immediate stimulation delivery.

This is a target-engagement and mechanistic-consistency hypothesis.

### H3: Route-participation breadth

After the routes have been independently validated, a noise-adjusted breadth statistic will be lower when one live route is selectively attenuated. Breadth will add held-out prediction of later visibility reports beyond total stimulus-linked transfer and competing neural or behavioral variables.

This is a predictive mechanistic hypothesis, not a direct causal consequence of H1.

### H4: Spectral distribution versus total transfer

A selective perturbation that changes the distribution of participation across routes will have a different association with reports from an empirically calibrated perturbation that produces a similar reduction in total route transfer while approximately preserving normalized route balance.

This is necessary before a causal role can be attributed specifically to breadth rather than to total transfer. It requires an additional successful manipulation and is not identified by statistical adjustment alone.

### H5: Nonlinear transition

A nonlinear breadth-by-strength model may predict visibility better than linear, monotonic-spline, and strength-only alternatives. A fitted midpoint will be described as a statistical transition only if observations span both sides, perturbations move participants across it, and the relation replicates.

This is a model-comparison hypothesis. It is no longer assumed that a neural “gate” exists.

### H6: Assumption-dependent mediation

Changes in neural route participation may statistically account for part of the randomized treatment effect under an explicit causal graph, joint measurement model, and sensitivity analysis.

This is not a prerequisite for H1 and will not be inferred from attenuation of a logistic-regression coefficient.

## 1.3 What the study can and cannot establish

If H1 is supported, the immediate conclusion is that the state of an unselected but task-admissible response system contributed to a later report. If H1 and H2 are both supported, the result is more specifically consistent with disruption of a prospectively live route. If H3 and H4 are also supported, balanced participation over validated routes becomes a plausible mechanistic feature.

Even that sequence would not establish:

- that phenomenal experience changed;
- that a report-derived endpoint directly measures consciousness;
- that route breadth is necessary or sufficient for conscious access;
- that a unique neural threshold exists;
- that speech-premotor or eye-field tissue is a selective output relay;
- that the result generalizes beyond this delayed, near-threshold task;
- that the same statistic applies to patients, infants, nonhuman animals, or artificial systems.

The terms **visibility report**, **confidence report**, and **flexible delayed reporting** are therefore preferred in results-facing statements. **Reportable access** is used only for the restricted theoretical construct being modeled, not as a synonym for phenomenality.

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# 2. Theoretical context

## 2.1 Predictive processing as a framework rather than a complete theory

Predictive processing has been proposed as a systematic basis for relating neural substrates to phenomenological structure and for organizing the search for neural correlates. Its usefulness partly follows from the fact that it is not, by itself, a complete theory of consciousness.[1] That distinction permits a constrained question about reporting architecture without claiming that predictive inference determines which states are conscious.

In the present task, sensory evidence can be represented hierarchically, maintained under masking and delay, and assigned different uncertainty. None of those properties alone determines whether the same content is prepared for manual, spoken, or saccadic use. The proposed experiment asks whether route structure contributes explanatory information after sensory evidence has been modeled.

Variational free-energy formulations may constrain mechanistic architectures without uniquely selecting one architecture, and the relationship between predictive processing and the free-energy principle remains debated.[28] The route-participation statistic introduced below is therefore an author-defined systems measure. It is not derived as a theorem of the free-energy principle.

## 2.2 Embodiment, precision, and potential confounding

Active interoceptive inference connects hierarchical prediction with autonomic regulation, uncertainty or precision, embodiment, emotion, and selfhood.[2] This literature motivates careful measurement of bodily and uncertainty-related variables, but it does not directly support the added claim that report-policy selection is governed by route breadth.

The distinction matters experimentally. Stimulation can change startle, discomfort, expectation, pupil-linked arousal, cardiac dynamics, respiration, or confidence. Those changes may affect delayed reports even if route structure is irrelevant. Emotion can influence perception, attention, action, learning, memory, reasoning, and executive control in context-dependent directions.[3] Physiological and subjective measurements are therefore competing explanatory variables rather than decorative controls.

## 2.3 Workspace theories and differentiated effects

Hierarchical thoughtseed proposals describe coherent embodied states that compete within a global-workspace framework and guide perception, action, and learning.[10] Information-theoretic workspace extensions have proposed phase-transition-like dynamics and sensitivity to broader biological and social context.[11] Integrated world-modeling theory similarly combines active inference, multimodal integration, attentional selection, and differentiated control within a theoretical synthesis.[34]

These are theoretical contexts, not empirical validation of the current task. In particular, they do not show that three report effectors constitute a consciousness mechanism or that perturbing an unused route changes experience.

A closer methodological antecedent is the Global Mediation Workspace proposal, which characterizes candidate mediating subnetworks through reachability, observability, input-output alignment, effective dimensionality, and routed source-target breadth.[17] That record describes a control-theoretic proposal, synthetic benchmarks, and a preliminary ECoG application in four macaques under ketamine. It does not validate trial-level human MEG estimates during online stimulation.

The present contribution is narrower than a general workspace criterion. It focuses on a randomized exclusion cue and asks whether a particular content-linked response route matters while it remains a valid option. The strongest plausible novelty relative to the stored records lies in that route-admissibility perturbation, not in the general use of spectral or control-theoretic language.

## 2.4 Why an unchosen route might matter

Before the final cue, a participant may need to preserve a content representation in a form usable by either of two valid response systems. Such preparation could involve:

- parallel route-specific transformations;
- a shared content code linked to multiple downstream controllers;
- maintenance of several response rules;
- verbal or spatial recoding;
- prospective attention to possible effectors;
- competition between response mappings;
- strategic postponement of commitment.

The original theory interpreted this structure as counterfactual policy reach. The revised proposal uses a more restrained formulation: the current neural and cognitive state may contain **prospective route-specific participation**. "Counterfactual" refers only to nearby randomized trials on which the alternative route would have been selected.

An unused route can matter physically without any unrealized future event acting backward in time. At the moment of stimulation, the participant has a current task set, learned mappings, route-specific neural activity, and an expectation that either remaining route may be required. The intervention changes that current state. The later final cue merely reveals that the targeted route will not be executed on the critical trial.

## 2.5 Why perturbation is needed

Correlations between later reports and frontal, parietal, premotor, or oculomotor activity can arise from attention, task rules, memory, decision, metacognition, or response preparation. Intracranial stimulation evidence also cautions against treating the prefrontal cortex as a unitary consciousness substrate. A synthesis of human intracranial electrical stimulation reported reliable experiential effects for some orbitofrontal and anterior-cingulate sites but comparatively few reportable alterations after anterolateral prefrontal stimulation.[35] That evidence neither validates the proposed TMS targets nor shows that absent elicited reports imply nonessentiality.

The relevant intervention is therefore not stimulation of an undifferentiated frontal region. It is stimulation of an independently localized response-associated system, crossed with whether that system is currently part of the valid task set.

## 2.6 Report measures remain reports

Binary visibility, ordered visibility, confidence, and meta-$d'$ are all generated through behavior. Preserved orientation discrimination does not prove that a selective visibility change reflects an earlier conscious state. It may reflect:

- a visibility criterion;
- confidence mapping;
- postdecision evidence;
- working-memory readout;
- demand characteristics;
- response formatting;
- a route-specific delay or conflict;
- altered willingness to endorse seeing.

The experimental sequence and models are revised to reduce avoidable contamination, but no report becomes a direct measure of phenomenality.

# 3. Operational concepts

## 3.1 Physical stimulus

The physical stimulus category is

$$U_i \in \{-1, +1\},$$

where $-1$ and $+1$ denote counterclockwise and clockwise orientation on trial $i$. Orientation assignment is randomized. Contrast and mask strength are separately recorded and, in designated conditions, randomized.

The intervention $\mathrm{do}(U_i = u)$ therefore has a clear experimental meaning: present physical orientation $u$. It does not mean setting a latent neural percept while holding the rest of the brain fixed.

## 3.2 Latent sensory evidence

Latent sensory evidence $E_i$ is a trial-specific variable inferred from stimulus properties, behavior, and sensory neural activity. It is endogenous to the participant's nervous system. It can mediate the effect of the randomized stimulus, but it is not itself randomized.

## 3.3 Decoded neural content

Decoded content $Z_{i,t}$ is an estimate obtained by applying an independently trained orientation decoder to neural data. It is a measurement or predictor, not an exogenous control input. Associations between $Z_{i,t}$ and later route coordinates may reflect common inputs, attention, arousal, leakage, or model structure.

## 3.4 Candidate report route

A candidate report route is a learned mapping from the orientation distinction to a route-specific output:

- manual keypress;
- spoken category;
- directed saccade.

The route label is provisional until perturbation localizers show that the associated response profile can be selectively changed without an equivalent change in all other candidates.

## 3.5 Validated route

A **validated route** is an equivalence class of candidate mappings supported by outcome-blind localizer evidence. Two candidates count as different validated routes only if their stimulus-linked outputs and perturbation-response profiles are sufficiently distinguishable under preregistered criteria.

Selective perturbability is thus used to validate the route partition. The later breadth statistic does not independently prove it.

## 3.6 Redundant mapping

Two mappings are redundant for the primary analysis when they differ in cue identity or superficial response rule but do not show distinct perturbation-response profiles. They are merged before route breadth is computed.

Three cue colors that all select the same manual controller do not create three routes merely because they have different labels. Conversely, multiple manual rules could remain distinct if an independent localizer showed reliably separable intermediate transformations and selective interventions. Effector identity is informative but not decisive.

### 3.7 Admissibility

A validated route $r$ is admissible at time $t$ when:

1. the instructions and cues received so far permit it to be selected by a later valid final cue;
2. its selection probability is nonzero;
3. the participant has learned the mapping;
4. route-readiness probes show that the route is psychologically live.

Admissibility is task-relative. Physical ability to speak does not make speech admissible after the cue has ruled it out.

### 3.8 Visibility report

The primary behavioral endpoint is a binary report answering whether the participant saw the orientation well enough to endorse its visibility under a standardized instruction. This is a report about visibility, not a direct observation of a conscious state.

An ordered visibility scale is collected in separate blocks rather than immediately after the binary judgment. The two measures are not repeated sequentially on the same trial in the confirmatory analysis.

### 3.9 Confidence and metacognitive statistics

Confidence is an ordered report about the participant's type-1 orientation choice. Meta-$d'$ is a model-dependent quantity inferred from type-1 choices and confidence under signal-detection assumptions. The ratio meta-$d'/d'$ is referred to as **metacognitive efficiency**, not objective metacognition. A selective change can reflect postdecision monitoring or confidence criteria and is not specific to conscious access.

### 3.10 Route-participation breadth

Route-participation breadth is the effective spectral breadth of stimulus-linked trajectories across a previously validated route set, evaluated in a fixed spatiotemporal noise geometry. It measures balanced, reliable participation over the route classes supplied to it.

It is not:

- the number of cue labels;
- ordinary matrix rank;
- independent controllability of the brain;
- causal independence discovered from observational data;
- total neural transfer;
- sensory evidence;
- a sentience score;
- a measure of phenomenal experience.

# 4. Fixed-space formal model

## 4.1 Overview

The revised formal object begins with fixed route labels and fixed trajectory coordinates. It does not sum matrices of changing dimensions, and it does not whiten each lag separately.

Let the union of candidate routes be

$$\mathcal{R} = \{\text{manual}, \text{speech}, \text{saccade}\}.$$

The output space is fixed throughout the analysis horizon, even when one route is excluded. Exclusion is represented by an admissibility mask, not by changing the output dimension.

Let the primary pre-final-cue analysis interval be

$$\mathcal{W} = [220, 360] \text{ ms}$$

relative to stimulus onset. This proposed window is a task hypothesis, not a known duration of consciousness. Theoretical work has emphasized that conscious contents have temporal extension and that neural dynamics may be content dependent, but it does not establish the present interval.[8]

The exclusion cue occurs before $\mathcal{W}$, and the final route-selection cue occurs after it. If pilot decoding does not show reliable task-set updating before the start of $\mathcal{W}$, the timing must be revised before preregistration.

## 4.2 Fixed trajectory basis

For route $r$, let $y_{ir}(\tau)$ be the route-specific orientation-decision coordinate on trial $i$ at time $\tau \in \mathcal{W}$. Positive values favor clockwise and negative values favor counterclockwise after participant-specific mapping alignment.

Rather than use every time sample, each trajectory is projected onto $L$ temporal basis functions fixed using treatment-blind localizer data:

$$\mathbf{y}_{ir} = \begin{bmatrix} y_{ir,1} & \cdots & y_{ir,L} \end{bmatrix}^\top \in \mathbb{R}^L.$$

The basis may consist of low-order splines or another preregistered smooth basis. Its dimension $L$, knot placement, and regularization are fixed before visibility outcomes are examined.

The full route-by-time output vector is

$$\mathbf{y}_i = \begin{bmatrix} \mathbf{y}_{i,\text{manual}}^\top & \mathbf{y}_{i,\text{speech}}^\top & \mathbf{y}_{i,\text{saccade}}^\top \end{bmatrix}^\top \in \mathbb{R}^q,$$

where

$$q = |\mathcal{R}|L.$$

The dimension $q$ is constant across lags, admissibility conditions, stimulation conditions, and participants within a fitted analysis.

## 4.3 Experimental condition

Let $c$ denote a condition defined by variables that are randomized or fixed before the trajectory is measured, including:

- exclusion-cue assignment;
- active or sham stimulation;
- stimulation time;
- mask or transfer-strength condition;
- route architecture block;
- participant and session strata.

The final cue lies outside $\mathcal{W}$. Critical manual-final trials can therefore be selected for the behavioral estimand without allowing the final response to determine the pre-final trajectory definition.

## 4.4 Average stimulus-linked response

For condition $c$, define the average signed stimulus-linked trajectory

$$\mathbf{k}_c = \frac{1}{2}\left[\mathbb{E}(\mathbf{y}_i \mid \mathrm{do}(U_i = +1), c) - \mathbb{E}(\mathbf{y}_i \mid \mathrm{do}(U_i = -1), c)\right].$$

Because physical orientation is randomized, this contrast can be interpreted as an average causal effect of physical orientation assignment on the measured route-output trajectory, conditional on the usual assumptions of consistency, no interference, correct randomization, and adequate measurement. It is not the causal effect of setting decoded content $Z$.

The same quantity can be estimated by randomized-condition regression, a state-space impulse-response model with explicit exogenous inputs, or another preregistered estimator. Agreement across estimators is a recovery criterion rather than an assumption.

## 4.5 Route-specific embedding

Let $S_r \in \mathbb{R}^{q\times q}$ be a fixed diagonal selection matrix retaining the $L$ coordinates belonging to route $r$ and setting all other coordinates to zero.

Define the embedded route response

$$\mathbf{d}_{r,c} = S_r\mathbf{k}_c.$$

Collect the route columns into

$$D_c = \begin{bmatrix}\mathbf{d}_{\mathrm{manual},c} & \mathbf{d}_{\mathrm{speech},c} & \mathbf{d}_{\mathrm{saccade},c}\end{bmatrix} \in \mathbb{R}^{q\times p},$$

where $p = |\mathcal{R}| = 3$.

The columns represent route-labeled trajectories. Their labels derive from the independently validated route partition. They are not eigenmodes inferred from $D_cD_c^\top$.

## 4.6 Admissibility mask

Let

$$\mathbf{a}_c = \begin{bmatrix} a_{\text{manual},c} & a_{\text{speech},c} & a_{\text{saccade},c} \end{bmatrix}^\top,$$

where $a_{r,c} = 1$ if route $r$ is admissible during the full interval $\mathcal{W}$ and $a_{r,c} = 0$ otherwise.

Define

$$\Pi_c = \operatorname{diag}(\mathbf{a}_c).$$

The output dimension remains fixed. Excluded routes contribute zero columns through $\Pi_c$.

If the admissible set changes inside $\mathcal{W}$, the trial is outside the primary model. The horizon is not allowed to cross the final cue or any other event that changes the policy set.

## 4.7 Fixed spatiotemporal covariance

Let

$$\Omega_0 \in \mathbb{R}^{q \times q}$$

be a full covariance matrix for content-independent residual variation in the route-by-time output space. It includes:

- covariance across routes;
- temporal covariance within a route;
- cross-route temporal covariance;
- source leakage represented in the measured coordinates.

The primary $\Omega_0$ is estimated from independent, treatment-blind baseline or sham-localizer data and is then held fixed across active, sham, admissible, excluded, and mask conditions. It is not re-estimated separately for each lag or treatment cell.

If regularization is required, the estimator and regularization strength are selected by nested validation on the independent data. If $\Omega_0$ is singular, analyses are restricted to its supported subspace using the Moore–Penrose pseudoinverse $\Omega_0^\dagger$.

Condition-specific residual covariance is estimated separately as a diagnostic. A treatment-induced noise change is not absorbed into the primary whitening metric.

## 4.8 Route-participation matrix

The revised route-participation matrix is

$$M_c = \Pi_c D_c^\top \Omega_0^\dagger D_c \Pi_c.$$

The matrix $M_c \in \mathbb{R}^{p \times p}$ is positive semidefinite because, for any vector $\mathbf{v} \in \mathbb{R}^p$,

$$\mathbf{v}^\top M_c \mathbf{v} = \left(D_c \Pi_c \mathbf{v}\right)^\top \Omega_0^\dagger \left(D_c \Pi_c \mathbf{v}\right) \geq 0.$$

Its diagonal entries quantify noise-adjusted route participation. Its off-diagonal entries reflect the geometry induced by the full inverse covariance.

This is not a standard finite-horizon controllability Gramian. Standard Gramian and rank criteria are established for particular control systems, including specialized stochastic mean-field systems, but those results do not validate this custom neural statistic.[18] Work on controllability-Gramian spectra and submatrices likewise concerns specified network-control models and does not show that entropy effective rank measures neural policy dimensions.[23][24]

The name **route-participation matrix** is used to prevent that overextension.

## 4.9 Total participation energy

Define total route participation as

$$T_c = \mathrm{tr}(M_c).$$

This is the overall noise-adjusted stimulus-linked energy represented across admissible validated routes.

A selective perturbation will often change both $T_c$ and spectral balance. These quantities are not experimentally separated merely by entering both into one regression.

## 4.10 Route-participation breadth

Let the eigenvalues of $M_c$ be

$$\lambda_{1,c}, \ldots, \lambda_{p,c},$$

with $\lambda_{j,c} \geq 0$. When $T_c > 0$, define

$$q_{j,c} = \frac{\lambda_{j,c}}{T_c}.$$

The route-participation breadth is

$$B_c = \exp\left(-\sum_{j=1}^{p} q_{j,c} \log q_{j,c}\right),$$

with $0 \log 0 = 0$.

This entropy effective rank is adopted here as a definition. It is not attributed to the stored control-theory sources.

If one eigenvalue is nonzero, $B_c = 1$. If $m$ nonzero eigenvalues are equal, $B_c = m$. Intermediate values express spectral balance.

At $T_c = 0$, breadth is undefined rather than set to zero. Arbitrarily weak but balanced responses can have high $B_c$, so breadth is never interpreted without total participation and uncertainty.

### 4.11 Continuous treatment of weak transfer

The primary randomized behavioral analysis includes all eligible trials and does not condition on a post-treatment threshold such as $T_c \geq T_{\min}$. Such conditioning could select on a mediator altered by stimulation.

Mechanistic models instead treat $T$ and $B$ jointly. A prespecified continuous reliability weight may be used:

$$w(T) = \frac{T}{T + T_0},$$

where $T_0 > 0$ is fixed using treatment-blind baseline data. A candidate behavioral model is

$$\operatorname{logit} \Pr(V_i = 1) = \alpha_s + f_T(\log T_i) + w(T_i) f_B(B_i) + \boldsymbol{\beta}^\top \mathbf{X}_i.$$

Here:

- $V_i$ is the binary visibility report;
- $\alpha_s$ is a participant-specific intercept;
- $f_T$ and $f_B$ are preregistered functions;
- $\mathbf{X}_i$ contains sensory, task, physiological, and report-process covariates.

The weighting prevents a high but unidentified breadth value at negligible signal from receiving the same interpretation as reliable breadth. It is a modeling choice and must be tested against unweighted and fully joint alternatives.

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## 5. Formal properties and construct checks

### 5.1 Invariance to a fixed invertible output transformation

Let the fixed trajectory coordinates be transformed by one invertible matrix

$$A \in \mathbb{R}^{q \times q},$$

so that

$$\mathbf{y}_i' = A \mathbf{y}_i.$$

Then

$$D_c' = A D_c$$

and

$$\Omega_0' = A \Omega_0 A^\top.$$

If $\Omega_0$ is positive definite,

$$\begin{aligned} M_c' &= \Pi_c D_c^\top A^\top \left(A\Omega_0 A^\top\right)^{-1} A D_c \Pi_c \\ &= \Pi_c D_c^\top \Omega_0^{-1} D_c \Pi_c \\ &= M_c. \end{aligned}$$

The route-participation matrix itself, not merely its eigenvalues, is unchanged.

This result depends on using a single fixed transformation and a single full covariance over the entire stacked trajectory. It does not apply to arbitrary lag-specific transformations paired with lag-specific whitening.

## 5.2 Exact measurement-duplication invariance

Suppose the trajectory is embedded into a taller measurement space by a full-column-rank matrix

$$L \in \mathbb{R}^{q' \times q}, \qquad q' \geq q.$$

Then

$$D_c' = L D_c$$

and

$$\Omega_0' = L \Omega_0 L^\top.$$

Using the pseudoinverse on the supported subspace,

$$L^\top \left(L\Omega_0 L^\top\right)^\dagger L = \Omega_0^{-1}$$

when $\Omega_0$ is positive definite and $L$ has full column rank. Therefore,

$$M_c' = M_c.$$

Exact duplication of measurement channels cannot increase breadth. This proof concerns exact linear measurement duplication under a fixed geometry. It does not show that approximately duplicated empirical channels with independent noise leave the estimate unchanged. Near-duplication remains a simulation and sensitivity problem.

## 5.3 Independent routes with identical time courses

Consider two validated routes with the same temporal response vector $\mathbf{g} \in \mathbb{R}^L$, equal amplitudes, and identity covariance. Their embedded columns occupy separate route blocks:

$$\mathbf{d}_1 = \begin{bmatrix} \mathbf{g} \\ \mathbf{0} \end{bmatrix}, \qquad \mathbf{d}_2 = \begin{bmatrix} \mathbf{0} \\ \mathbf{g} \end{bmatrix}.$$

Then

$$D^\top D = \|\mathbf{g}\|^2 I_2.$$

The two eigenvalues are equal, and $B = 2$.

The statistic therefore does not collapse independently validated routes merely because their temporal profiles are proportional.

### 5.4 Temporal change within one route

Suppose one validated route has a trajectory that changes direction or basis loading over time. It still contributes one route column $\mathbf{d}_1$. With only one validated admissible route,

$$M = \mathbf{d}_1^\top \Omega_0^\dagger \mathbf{d}_1$$

is one-dimensional and $B = 1$ whenever the response is nonzero.

Temporal rotation within one route cannot create extra route breadth.

### 5.5 Common drive

Suppose a common latent source produces similar activity in two candidate output coordinates, but no selective perturbation can alter one without equivalently altering the other. The breadth matrix alone may still assign participation to both labeled columns. It cannot determine that the two candidates are causally redundant.

For that reason, the route partition must be fixed by independent intervention evidence. If the common-drive candidates fail intervention separability, they are merged or declared unresolved. Computing breadth over them as though they were independent is prohibited.

This is a construct limitation, not an estimator detail.

### 5.6 Redundant cue mappings

If three cue colors all select mappings implemented by one validated manual route, they enter one route class. Their cue identities may be represented in task-set models, but they do not create three columns in $D_c$.

If a supposedly redundant mapping produces a distinct selective-intervention profile, it cannot be treated as redundant merely to preserve the theory. The route partition must be revised before outcome analysis.

### 5.7 Correlated route noise

When route outputs have correlated noise, $\Omega_0$ is not block diagonal. The resulting off-diagonal structure in $M_c$ affects its eigenvalues. This is intentional: two routes whose signals are difficult to distinguish relative to correlated noise should not be treated as equivalent to two independent, equally reliable measurements.

The analysis nevertheless distinguishes signal and noise. The primary metric uses one reference covariance, while condition-specific covariance changes are reported separately. A stimulation-induced increase in route noise is not automatically interpreted as reduced signal participation.

### 5.8 Selective attenuation

Let the baseline route-participation matrix be diagonal with $N$ equal nonzero entries. If one route's response amplitude is multiplied by $\rho$, with $0 \leq \rho \leq 1$, its participation energy is multiplied by $\rho^2$.

The normalized energies are then

$$\frac{1}{N-1+\rho^2}$$

for each unaffected route and

$$\frac{\rho^2}{N-1+\rho^2}$$

for the target route. Breadth becomes

$$B_N(\rho) = \exp\left[\log(N-1+\rho^2) - \frac{\rho^2}{N-1+\rho^2}\log\rho^2\right].$$

This expression is valid only for the idealized diagonal, equal-energy case. It is not used to infer that empirical route columns coincide with eigenmodes.

### 5.9 Proportional attenuation

If all route trajectories are multiplied by a common amplitude $a > 0$,

$$D'_c = aD_c,$$

then

$$M'_c = a^2 M_c.$$

Total participation changes as

$$T'_c = a^2 T_c,$$

while normalized eigenvalues and breadth remain unchanged:

$$B'_c = B_c.$$

This formal separation motivates an experimental comparison between selective attenuation and approximately proportional attenuation. It does not guarantee that any proposed biological manipulation will achieve the required proportionality.

---

## 6. Interventional validation of route classes

### 6.1 Independent localizer requirement

Route classes are established in sessions whose data are separated from the primary visibility analysis. Visibility outcomes from the principal task cannot be used to decide:

- which route outputs are retained;
- which mappings are merged;

- which target is “selective”;
- which temporal basis is used;
- which covariance regularization is chosen;
- which participants count as target engagers.

This separation prevents the outcome effect from defining the mechanism that is later invoked to explain it.

## 6.2 Localizer design

Participants perform high-visibility orientation reports using each candidate route. Active and sham perturbations are randomized within the localizer.

For perturbation axis $j$ and route $r$, define a route-response contrast

$$\Psi_{rj} = \mathbb{E}(A_r \mid \mathrm{do}(P_j = 1)) - \mathbb{E}(A_r \mid \mathrm{do}(P_j = 0)),$$

where:

- $P_j = 1$ denotes active perturbation on axis $j$;
- $P_j = 0$ denotes sham or matched control;
- $A_r$ is a prespecified route-specific neural or behavioral response measure.

The matrix

$$\Psi = [\Psi_{rj}]$$

is the selective-intervention response matrix. Its columns describe how each perturbation changes the candidate routes.

## 6.3 Route-separability criteria

A candidate route is retained as distinct only if all of the following are satisfied in held-out localizer data:

1. its orientation decision coordinate is reliable across runs;
2. selected-route behavior shows nontrivial route-specific performance;
3. the intended perturbation changes the target route by at least a prespecified smallest effect of interest;
4. changes in non-target routes fall within prespecified spillover bounds or are substantially smaller than the target change;
5. the perturbation profile is not statistically equivalent to that of another candidate route;
6. the result replicates across independent localizer runs.

Provisional planning values may use a target-route standardized effect of $0.20$ and non-target equivalence bounds of $\pm 0.10$, but these are not claimed to be validated. Final values must be chosen from blinded pilot operating characteristics and published before confirmatory recruitment.

### 6.4 Outcome-independent route quotient

Candidate mappings are represented as nodes in a graph. Two nodes are merged when their perturbation-response vectors, route-output trajectories, and selected-route behavioral consequences are equivalent within preregistered bounds. Connected components define route classes.

This graph and its equivalence thresholds are frozen before the principal visibility outcome is analyzed.

If a speech and manual mapping cannot be selectively separated, the analysis does not label them as two routes. If three manual remappings unexpectedly show separable intervention profiles, they are not forced into one route for theoretical convenience.

### 6.5 Population and participant-level target engagement

The primary randomized analysis follows an intention-to-treat principle for all eligible participants. A target-engager subgroup may be analyzed only if engagement was classified from an independent session before principal-task outcomes were available.

Selection of participants based on post-randomization visibility effects is prohibited. The target-engager analysis estimates an effect in a selected, explicitly defined population and cannot be generalized automatically to all participants.

### 6.6 Equal perturbation delivery across admissibility states

A smaller behavioral effect after route exclusion may occur because the stimulated tissue or network responds less strongly in that state. The experiment therefore measures:

- pulse intensity and coil placement;
- immediate artifact-corrected local response where technically recoverable;
- broader evoked network response;
- auditory and somatic responses;
- route-coordinate suppression;
- state-dependent baseline activity.

Physical delivery must be comparable across admissible and excluded conditions. If the early local neural response is materially weaker after exclusion, the interaction cannot uniquely distinguish route irrelevance from state-dependent stimulation gain. That result remains informative about task-state dependence but does not identify the proposed route mechanism.

---

## 7. Exogenous inputs and dynamic estimation

### 7.1 Explicit state-space inputs

A state-space estimator may be used, but it must include the experimentally assigned events explicitly. Let

$$x_{i,t} \in \mathbb{R}^n$$

be a latent neural state. A local model is

$$x_{i,t+1} = A_t x_{i,t} + B_t^U U_i h_t^U + B_t^G G_i h_t^G + B_t^A \mathbf{a}_i h_t^A + B_t^P P_i h_t^P + B_t^F F_i h_t^F + \Gamma_t \mathbf{L}_i + \xi_{i,t}.$$

Here:

- $U_i$ is randomized physical orientation;
- $G_i$ is randomized mask or transfer-strength condition;
- $\mathbf{a}_i$ encodes the exclusion-cue assignment;
- $P_i$ encodes active or sham perturbation and target;
- $F_i$ encodes the final cue;
- $h_t^{\cdot}$ are fixed event-time functions;
- $\mathbf{L}_i$ contains measured pretreatment covariates;
- $\xi_{i,t}$ is process noise.

The final-cue term is included in the whole-trial model even though the primary route-breadth horizon ends before that cue.

Observed neural data satisfy

$$o_{i,t} = H_t x_{i,t} + \nu_{i,t},$$

with measurement noise $\nu_{i,t}$.

Interactions such as perturbation by admissibility must be modeled explicitly if the estimator is used to recover condition-specific trajectories.

### 7.2 Predictive versus causal transfer

Impulse responses from randomized physical stimulus input $U_i$ can be interpreted as stimulus-assignment effects if the model is correctly specified. Impulse responses from decoded content $Z_{i,t}$ remain predictive because $Z$ is an endogenous neural estimate.

The manuscript therefore uses:

- **stimulus-linked route response** for effects anchored to randomized $U$;
- **neural-content predictive transfer** for associations anchored to $Z$;
- **route perturbation effect** for changes caused by randomized stimulation.

The term **causal content reach** is not used.

### 7.3 Similarity invariance and its limits

For a fixed invertible latent-state transformation $T_x$,

$$x'_{i,t} = T_x x_{i,t},$$

the transformed state matrices can represent the same input-output process. Exact stimulus-to-output impulse responses are invariant under the corresponding similarity transformation.

This coordinate invariance does not establish:

- minimality;
- observability;
- sufficient excitation;
- causal meaning for every latent state;
- absence of latent common input;
- correct measurement-error specification;
- uniqueness among observationally equivalent nonminimal models.

The state-space realization is therefore one estimator of the fixed-space route trajectories, not the definition of the construct.

### 7.4 Identification assumptions

Interpretation of dynamic estimates requires explicit assumptions:

1. randomized exogenous inputs are encoded correctly;
2. model order is sufficient but not gratuitously nonminimal;
3. the measured system is adequately observable over the chosen horizon;
4. the randomized inputs provide sufficient excitation;
5. unmeasured common causes do not generate the entire estimated route response;
6. source leakage is represented adequately in the observation model or covariance;
7. measurement error does not depend on visibility in an unmodeled way;
8. TMS artifacts are separable from neural response in the retained interval;
9. the route-output coordinates remain stable across sessions;
10. alternative state-space realizations that fit measured covariance yield materially similar route-participation estimates.

Violation of these assumptions may invalidate H2–H5 while leaving the randomized behavioral H1 estimable.

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## 8. Causal estimands

### 8.1 Variables

Let:

- $T_i \in \{0, 1\}$ denote sham versus active route-targeted stimulation;
- $A_i \in \{0, 1\}$ denote target excluded versus target admissible;
- $G_i \in \{0, 1\}$ denote standard versus global transfer-attenuation condition;
- $Y_i \in \{0, 1\}$ denote the binary visibility report;
- $D_i$ denote type-1 orientation discrimination;
- $N_i$ denote post-treatment neural variables, including target-route participation, total participation $T_i^{\text{neural}}$, and breadth $B_i$;

- $Q_i$ denote post-treatment arousal, discomfort, expectancy, task-set, memory, and response-process variables;
- $L_i$ denote pretreatment covariates.

Neural total participation is written $T_i^{\text{neural}}$ where necessary to distinguish it from treatment $T_i$.

## 8.2 Primary total-effect interaction

Let $Y_i(t, a, g)$ be the potential visibility report under treatment $t$, admissibility condition $a$, and transfer-strength condition $g$.

The primary estimand at the standard transfer condition is the marginal risk-difference interaction

$$\Delta_Y = [\mathbb{E}\{Y(1,1,0)\} - \mathbb{E}\{Y(0,1,0)\}] - [\mathbb{E}\{Y(1,0,0)\} - \mathbb{E}\{Y(0,0,0)\}] \,.$$

The hypothesis predicts $\Delta_Y < 0$: active stimulation lowers the probability of endorsing visibility more when the target route is admissible.

Marginal risk differences are primary because logistic coefficients are noncollapsible. Odds-ratio interactions are reported secondarily.

## 8.3 Neural treatment effects

Corresponding neural estimands are defined for target-route participation, total participation, and breadth:

$$\Delta_N = [\mathbb{E}\{N(1,1,0)\} - \mathbb{E}\{N(0,1,0)\}] - [\mathbb{E}\{N(1,0,0)\} - \mathbb{E}\{N(0,0,0)\}] \,.$$

For target-route participation and breadth, the predicted sign is negative. No sign is imposed on all other neural variables.

## 8.4 Predictive neural association

The trial-level neural association asks whether posterior route breadth predicts $Y_i$ after accounting for treatment, admissibility, stimulus evidence, total participation, task-set decoding, physiological state, and report order.

This association is not a randomized mediator effect. It is reported separately from the total treatment effect.

## 8.5 Assumption-dependent interventional indirect effect

A stochastic interventional indirect effect may be estimated by comparing expected reports when the neural mediator distribution is shifted from its sham distribution to its active-stimulation distribution while treatment is otherwise held fixed.

For admissibility state $a$, a schematic estimand is

$$\mathrm{IIE}_a = \mathbb{E}\left[Y\left(1, a, N \sim F_{N|T=1,A=a,L,Q}\right)\right] - \mathbb{E}\left[Y\left(1, a, N \sim F_{N|T=0,A=a,L,Q}\right)\right] \,.$$

Estimation requires:

- consistency;
- positivity;

- correct measurement of the neural mediator;
- correct models for post-treatment variables;
- no unmeasured mediator–outcome confounding after conditioning;
- a defensible treatment of exposure-induced confounders $Q$;
- no interference;
- correct temporal ordering.

These assumptions are strong. Stimulation may create arousal, discomfort, response slowing, or local evoked activity that affects both estimated breadth and reporting. Mediation is consequently a sensitivity analysis, not a decisive criterion. Coefficient attenuation after adding $B_i$ is not accepted as evidence of mediation.

### 8.6 Causal graph

The preregistered graph contains the following directed relations:

- $U \to E \to N \to Y$;
- $U \to D$ and $E \to D$;
- $A \to N$ and $A \to Q$;
- $T \to N, T \to Q$, and potentially $T \to Y$;
- $G \to E, G \to N$, and $G \to Y$;
- $L \to E, N, Q, Y$;
- $Q \to N$ and $Q \to Y$.

Alternative graphs add direct effects from task-set maintenance, memory, criterion setting, and response formatting. Model comparisons and sensitivity analyses evaluate those alternatives; no graph is declared true by stipulation.

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## 9. Separating breadth from total transfer

### 9.1 Why regression adjustment is insufficient

Selective attenuation of one route will generally reduce both total participation and breadth. Since both are estimated from the same matrix, they will be correlated and share measurement error. Entering them into one regression does not create experimental orthogonality.

The revised program therefore separates two questions:

1. Does an admissible unchosen route affect later reporting?
2. Is spectral breadth, rather than total route transfer or another route-state variable, the relevant mechanism?

The main single-site experiment can answer the first under successful target validation. It cannot alone answer the second.

## 9.2 Global transfer-attenuation comparator

A randomized mask-strength or contrast manipulation is proposed as an initial comparator. In a separate blinded calibration sample, a stronger mask level would be selected to:

- reduce total route participation by an amount comparable to selective route stimulation;
- reduce early sensory evidence in a measured and modeled way;
- leave normalized route-participation breadth approximately unchanged.

This manipulation is imperfect because it acts upstream on sensory evidence. It can demonstrate that equal reductions in total participation need not produce equal spectral changes, but it cannot by itself isolate breadth from all differences between sensory and route-specific perturbations.

## 9.3 Balanced-route attenuation comparator

A stronger mechanistic stage would use calibrated perturbation of both currently admissible route systems, with each route attenuated by approximately the same proportion. In the manual-plus-speech condition, for example, paired interventions would target the manual and speech-associated systems. The intended result is a substantial reduction in total participation with minimal change in normalized breadth.

This design is technically and ethically demanding. It introduces extra stimulation, potential direct impairment of the final manual response, site-dependent sensations, and greater artifact burden. It is therefore not part of the initial confirmatory H1 experiment. It proceeds only if separate pilots establish:

- safe and tolerable delivery;
- approximately proportional neural attenuation;
- matched auditory and somatic effects;
- interpretable manual behavior;
- adequate trial yield;
- no unacceptable broad sensory disruption.

If neither the global nor balanced comparator achieves trace matching and breadth preservation, H4 remains untested. The manuscript will not infer breadth causality from H1–H3 alone.

## 9.4 Matched-effect analysis

When a valid comparator exists, conditions are compared in regions of overlapping posterior total participation. Matching or weighting is based on the randomized manipulation and pretreatment variables, not on excluding post-treatment trials solely because their observed total participation misses a target.

The decisive pattern would be:

- similar change in total participation;
- different change in breadth;
- corresponding difference in visibility reports;
- preserved interpretation after sensory evidence and nonspecific treatment effects are modeled.

Even this pattern would remain contingent on the comparator’s exclusion assumptions.

# 10. Simulation and pilot program

## 10.1 Stage-gated status

The proposal is not ready for confirmatory implementation. Before human hypothesis testing, simulations and treatment-blind pilots must establish that the estimator and interventions can recover the intended quantities.

The program has five stages:

1. analytic and synthetic construct validation;
2. realistic neural-estimator recovery;
3. behavioral and route-readiness pilot;
4. stimulation, artifact, and target-engagement pilot;
5. confirmatory randomized experiment.

Failure at an earlier stage prevents advancement.

## 10.2 Analytic construct suite

The implementation must reproduce the following known cases:

| Generative case | Required result |
|---|---|
| Two selectively defined routes with identical temporal profiles | Breadth approaches 2 when energies and noise are balanced |
| One route with changing temporal profile | Breadth remains 1 |
| Exact duplicate measurement channels | $M$, total participation, and breadth remain unchanged |
| Duplicate cue labels mapped to one route class | Route count and breadth do not increase |
| Proportional attenuation of all routes | Total participation falls; breadth is unchanged |
| Selective attenuation of one balanced route | Both total participation and breadth fall |
| Fixed invertible output transformation | $M$ is unchanged |
| Lag-dependent output rotations analyzed incorrectly | Recovery test demonstrates why the withdrawn estimator fails |

These tests are unit tests of the definition, not evidence about brains.

## 10.3 Generative simulation families

Synthetic recovery must include at least the following families:

1. independently perturbable routes with identical timing;
2. redundant outputs with different apparent timing;
3. one shared route with temporally rotating measurement loadings;
4. shared-hub and common-drive architectures;
5. independent routes with correlated output noise;
6. route-specific and global noise changes caused by stimulation;

7. exact and near-duplicate measurement channels;
8. source leakage between adjacent nodes;
9. latent common inputs;
10. model-order underfitting and overfitting;
11. nonlinear route interactions;
12. trial-varying route gains;
13. TMS artifacts with missing or unrecoverable intervals;
14. condition-dependent trial rejection;
15. criterion shifts with no neural breadth effect;
16. sensory-only treatment effects;
17. total-transfer-only effects;
18. task-set effects with no route-specific transfer;
19. route-specific transfer effects with no visibility effect;
20. complete null data.

## 10.4 Recovery metrics

Before confirmatory use, the pipeline must meet preregistered criteria for:

- bias in total participation;
- bias in breadth;
- interval coverage;
- route-classification error;
- false detection of extra routes;
- false-positive H1 interaction;
- false-positive breadth increment;
- sensitivity to condition-dependent covariance;
- calibration of equivalence decisions;
- held-out prediction;
- robustness across plausible source leakage and model orders.

Provisional design goals are:

- median absolute breadth error below $0.10$ on the two-route scale;
- at least 85% coverage for nominal 90% intervals across the prespecified model family;
- false-positive probability no greater than 0.05 for the primary decision rule under each core null family;
- at least 90% simulated detection probability at the final smallest effect of interest;
- no more than 5% false splitting of exact or near-redundant routes under the accepted noise regime.

These values are planning targets, not achieved results.

## 10.5 Reliability pilot

A treatment-blind pilot must estimate:

- within-participant reliability of each route coordinate;
- stability of the temporal basis;
- stability of $\Omega_0$;
- reliability of total participation;
- reliability of breadth;
- route-readiness decoding;
- usable trial fraction;
- between-session transfer of localizers;
- number of trials required per factorial cell.

Breadth is not advanced to a trial-level predictor unless its posterior reliability exceeds a preregistered threshold and simulation shows that apparent trial variability is not dominated by estimation noise.

## 10.6 Stimulation and artifact pilot

The stimulation pilot must establish:

- route-specific selected-response impairment;
- limited non-target impairment;
- equal physical delivery across admissibility states;
- recoverable neural data in the prespecified post-pulse interval;
- auditory, muscle, blink, and eye-movement artifact characterization;
- condition-independent rejection rates or adequate correction;
- acceptable fatigue and adverse-event rates;
- feasible session duration.

No stored source record supplied here establishes these capabilities. Their inclusion is therefore a required empirical gate rather than an assumed method property.

## 10.7 Trial-count simulation

Participant count alone is not an adequate power argument. The final simulation must specify:

- usable trials per active-by-admissibility cell;
- route-final probabilities;
- stimulation-time allocation;
- expected artifact loss;
- missed-response loss;
- route-readiness failures;
- localizer reliability;
- repeated-session correlation;
- meta-$d'$ precision;
- attrition.

If the required number of trials exceeds tolerable session burden, the design must be simplified rather than relying on nominal recruitment.

# 11. Experimental design

## 11.1 Participants and samples

The program proposes:

- one observational MEG development cohort;
- one speech-target perturbation cohort;
- one oculomotor-target perturbation cohort;
- an independent replication for any effect promoted as confirmatory.

A provisional target is 96 usable participants per primary perturbation cohort, but this number has no evidential status until the trial-level simulation is complete. Recruitment targets must account for target-localizer failure, artifact loss, attrition, and the distinction between intention-to-treat and target-engager analyses.

## 11.2 Stimulus

The primary stimulus is a briefly presented, masked Gabor patch tilted clockwise or counterclockwise. Contrast is calibrated to produce above-chance discrimination and a broad distribution of visibility reports without concentrating all trials at floor or ceiling.

The staircase is run before confirmatory blocks. Any online updating follows a fixed rule blind to stimulation and admissibility condition. Physical contrast sequences are yoked across key conditions where feasible.

## 11.3 Report routes

Participants learn three mappings:

- **manual:** one of two keys;
- **spoken:** one of two trained category words;
- **saccadic:** one of two response targets.

Mapping direction is counterbalanced. Route-specific performance is trained to a prespecified criterion before the participant enters the principal task.

## 11.4 Trial timeline

A provisional trial timeline is:

| Time from stimulus onset | Event |
|---|---|
| $-800$ to $-300$ ms | Fixation and block-context reminder |
| $0$ ms | Near-threshold orientation onset |
| $33$ ms | Stimulus offset |
| $50$–$100$ ms | Pattern mask |
| $140$ ms | Route-exclusion cue |

| Time from stimulus onset | Event |
| --- | --- |
| 220–360 ms | Primary route-participation interval |
| 260 or 340 ms | Candidate post-exclusion stimulation center |
| 700 ms | Final route-selection cue |
| after final cue | Report sequence determined by block |
| end of sampled trials | Sensation, expectancy, or interoceptive probes |

A 130 ms stimulation condition occurs before the exclusion cue. It is explicitly a pre-exclusion negative-control condition, not a test of current admissibility. Because the future exclusion assignment is randomized and not yet revealed, a 130 ms interaction would suggest imbalance, anticipation, sensory disruption, or analysis error.

## 11.5 Exclusion-cue conditions

For speech-target sessions:

- **target admissible:** saccade is excluded, leaving manual and speech;
- **target excluded:** speech is excluded, leaving manual and saccade.

For oculomotor-target sessions:

- **target admissible:** speech is excluded, leaving manual and saccade;
- **target excluded:** saccade is excluded, leaving manual and speech.

The final cue selects one of the remaining routes with equal probability in the primary design. Manual-final trials form the critical behavioral comparison. Target-final trials validate selected-route efficacy.

## 11.6 Cue matching

The exclusion cues are matched for:

- visual discriminability;
- frequency;
- information;
- expected reward;
- time to final cue;
- number of remaining routes;
- response deadline.

Subjective cue interpretation is assessed in separate probe trials. Route identity, rather than nominal cue entropy, is the intended difference.

## 11.7 Verification of task-set updating

A decoder trained independently on cue and route-readiness activity must show that the participant's task state distinguishes target-admissible from target-excluded conditions before the post-exclusion stimulation center.

If updating occurs too late, the stimulation time is moved in the pilot phase and then frozen. A confirmatory analysis may not redefine the window after examining visibility effects.

### 11.8 Report-order design

The original sequence placed binary visibility, confidence, and ordered visibility after the type-1 choice, creating anchoring and consistency effects. The revised design uses separate blocks.

#### Visibility-first blocks

After the final route cue, the participant first makes a binary visibility report using a fixed response format not involving the targeted speech or saccadic route. The selected orientation report follows.

Binary visibility in these blocks is the primary behavioral endpoint. It occurs after final route selection but before the type-1 choice, reducing contamination by correctness feedback, motor success, and postdecision confidence.

#### Choice-first blocks

The participant first makes the selected orientation report and then gives confidence. These blocks estimate type-1 sensitivity and metacognitive statistics under the conventional order.

#### Ordered-visibility blocks

A four-level visibility report replaces the binary report. The two scales are not collected sequentially on the same trial. Ordered visibility is a secondary outcome and an order-control analysis.

No ordering scheme turns visibility into a direct measure of phenomenality. The split design only makes specific postdecision alternatives more testable.

### 11.9 Provisional trial burden

A simplified primary stimulation session crosses:

- active versus sham;
- target admissible versus target excluded;
- standard versus global transfer-attenuation condition.

This yields eight principal manual-final cells. A provisional minimum of 48 usable trials per cell would require 384 critical trials, plus target-final validation and control trials. Trials would be distributed across multiple sessions with prespecified fatigue limits.

Chronometry conditions are evaluated in a separate pilot or subsample rather than fully crossing four stimulation times with every confirmatory factor. The final count is determined by simulation, not by the provisional values above.

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## 12. Neural measurements and route estimation

### 12.1 Observational MEG

Whole-head MEG is proposed for the development cohort, with structural imaging for individualized source modeling and simultaneous recording of:

- eye position;
- blinks;
- facial and limb electromyography;
- audio;
- pupil diameter;
- electrocardiography;
- respiration;
- head position.

The manuscript does not assume that source-resolved MEG can reliably recover the proposed trial-level statistic. That is a pilot question.

## 12.2 Sensory localizer

Independent high-contrast trials train an orientation decoder. The confirmatory early sensory interval is restricted to a cue-free period, provisionally $60–120$ ms after stimulus onset. This avoids contamination by the exclusion cue at 140 ms.

Later sensory persistence is modeled separately and includes cue-evoked inputs explicitly.

## 12.3 Route localizers

High-visibility trials identify manual, spoken, and saccadic orientation-decision coordinates. Localizer outputs are aligned so that their signs refer to the same physical category.

A route coordinate is retained only if it is:

- reliable across runs;
- predictive of the selected route's category;
- measurable before overt movement in the relevant interval;
- distinguishable from artifact;
- compatible with selective-intervention validation.

## 12.4 Source uncertainty

The proposed source architecture includes occipitotemporal sensory systems, parietal and frontal task systems, hand motor planning, speech-associated premotor systems, and oculomotor planning systems. These are candidate measurement domains, not consciousness centers.

Deep-source estimates, including thalamic estimates, are exploratory. The route-breadth test does not depend on successful thalamic localization.

## 12.5 Concurrent stimulation electrophysiology

TMS-compatible electrophysiology may be collected in perturbation sessions. The primary neural interval must exclude pulse-contaminated samples unless pilot evidence shows recoverable signal under a frozen pipeline.

Artifact processing is performed blind to visibility outcome. Condition-dependent data loss is reported. If active and sham conditions differ materially in recoverable samples, neural mediation is not interpreted without sensitivity bounds.

### 12.6 Signal and noise decomposition

For each condition, the analysis reports separately:

- stimulus-linked mean trajectory $\mathbf{k}_c$;
- fixed-reference route-participation matrix $M_c$;
- total participation $T_c$;
- breadth $B_c$;
- condition-specific residual covariance;
- target-route response;
- non-target route responses;
- task-set decoding;
- scalar amplitude and spatial spread;
- early sensory decoding.

A breadth change produced primarily by a condition-specific noise alteration receives a different interpretation from a change in the stimulus-linked response profile.

### 12.7 Alternative estimators

At least two estimators are compared:

1. a randomized-condition regression in the fixed trajectory basis;
2. a dynamic state-space estimator with explicit exogenous inputs.

A third nonlinear or finite-amplitude estimator may be included if simulations justify it.

The statistic is considered recoverable only if substantively equivalent conclusions arise across estimators within prespecified tolerance. If observationally equivalent state-space fits yield unstable breadth, the trial-level mechanism claim is abandoned.

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## 13. Stimulation design

### 13.1 Individualized targets

Speech-associated and oculomotor-associated targets are individualized using structural anatomy, route localizers, and connectivity estimates. The target must lie in a system whose localizer response is linked to the designated route.

The target is not assumed to be a selective output relay. Speech-associated systems may contribute to verbal recoding, working memory, or task rules. Oculomotor-associated systems may contribute to spatial attention or visual selection. These alternatives are part of the causal interpretation.

## 13.2 Active, sham, and control conditions

The principal comparison uses active and sham stimulation matched as closely as feasible for acoustic and somatic experience. A control site may be added if pilot data show that it provides a credible sensory match without affecting route-specific measures.

The primary causal estimand remains active versus sham crossed with admissibility. A target-versus-control-site contrast is supportive but does not replace sham matching.

## 13.3 Chronometry

Provisional stimulation centers are:

- 130 ms: pre-exclusion control;
- 260 ms: post-exclusion early route-state test;
- 340 ms: post-exclusion late route-state test;
- 450 ms: later pre-final-cue control.

The confirmatory experiment uses one post-exclusion center selected from pilot evidence and frozen before outcome collection. Chronometric claims require replication at the same time.

## 13.4 Selected-route validation

When the final cue selects the target route, active stimulation should alter a prespecified route-specific endpoint:

- spoken report accuracy or onset latency;
- saccade direction, endpoint, or latency;
- route-specific neural response.

Selected-route impairment shows that the intervention affects behavior associated with the target system. It does not prove selectivity or demonstrate the unchosen-route hypothesis.

## 13.5 Critical manual-final trials

On critical trials, the final orientation response is manual regardless of whether the target speech or saccadic route was previously admissible. The target route is not executed.

The central contrast therefore cannot be explained by direct failure to articulate or direct failure to make the final saccade. It can still be explained by route-specific task-set, memory, attention, confidence, or report-formatting processes.

## 13.6 Failure criteria for selectivity

The route-specific interpretation fails if active stimulation produces any of the following beyond prespecified bounds:

- comparable impairment of the manual final response;
- comparable loss of early sensory decoding;
- broad disruption of all route coordinates;
- large change in task-set decoding unrelated to target admissibility;
- large condition-specific arousal or discomfort effect;

- unrecoverable stimulation artifacts concentrated in one condition;
- substantially weaker local perturbation delivery after route exclusion.

Such a failure does not necessarily invalidate the randomized total effect. It invalidates the selective route mechanism.

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# 14. Behavioral and physiological model

## 14.1 Type-1 evidence

A hierarchical equal-variance signal-detection model uses

$$E_i \sim \mathcal{N}(\mu_{is} U_i, 1),$$

where $\mu_{is} > 0$ may vary with participant $s$, contrast, mask condition, fatigue, cue condition, and stimulation.

Condition-specific type-1 sensitivity is

$$d'_{is} = 2\mu_{is}$$

under this parameterization. The notation is condition specific because the design allows sensitivity to vary.

The type-1 choice depends on $E_i + b_s$, where $b_s$ is a participant-specific category bias.

## 14.2 Visibility model

The primary model estimates the marginal treatment-by-admissibility effect without requiring a neural mediator:

$$\operatorname{logit} \Pr(Y_i = 1) = \alpha_s + \beta_T T_i + \beta_A A_i + \beta_{TA} T_i A_i + \beta_G G_i + \boldsymbol{\gamma}^\top \mathbf{L}_i.$$

The reported primary quantity is the standardized marginal risk-difference interaction derived from this model.

Neural variables are introduced only in secondary models.

## 14.3 Joint breadth-and-strength model

The mechanistic predictive model is

$$\operatorname{logit} \Pr(Y_i = 1) = \alpha_s + \beta_T T_i + \beta_A A_i + \beta_{TA} T_i A_i + f_E(|E_i|) + f_T(\log T_i^{\text{neural}}) + w(T_i^{\text{neural}}) f_i$$

Posterior uncertainty in $E_i$, $T_i^{\text{neural}}$, and $B_i$ is propagated jointly. Plug-in point estimates are not treated as error free.

## 14.4 Threshold alternatives

Four relationships between breadth and visibility are compared:

1. linear;

2. monotonic spline;
3. smooth sigmoid;
4. change-point or threshold model.

A participant-specific sigmoid is

$$\Pr(Y_i = 1) = \sigma\left[k_s(B_i - \theta_s) + f_T(\log T_i^{\mathrm{neural}}) + \mathbf{c}_i^\top \boldsymbol{\beta}\right].$$

The midpoint $\theta_s$ is not called a neural gate unless:

- data occupy both sides of the fitted midpoint;
- perturbation moves observations across it;
- the threshold model improves held-out prediction by the preregistered margin;
- the result replicates across route identities or an independent cohort;
- linear and monotonic alternatives are inadequate.

## 14.5 Confidence and meta-$d'$

Confidence is modeled ordinally after the type-1 report in choice-first blocks. Meta-$d'$ and $d'$ are estimated jointly in a hierarchical signal-detection model.

The primary metacognitive quantities are:

- meta-$d'$;
- $d'$;
- meta-$d' - d'$ on the latent scale;
- meta-$d'/d'$ where the denominator is sufficiently identified.

A change in metacognitive efficiency does not distinguish earlier visibility from postdecision monitoring without additional evidence.

## 14.6 Response time

Type-1, visibility, and confidence response times are modeled separately. A generic model is

$$\log \tau_i \sim \mathcal{N}\left(\nu_s + g_E(|E_i|) + g_T T_i + g_A A_i + g_{TA} T_i A_i + g_N(N_i), \omega_s^2\right).$$

An apparent visibility effect explained by missed deadlines, motor slowing, or longer deliberation counts against a route-breadth interpretation.

## 14.7 Criteria and report formatting

Binary and ordinal visibility models include condition-specific thresholds. The analysis tests whether active stimulation changes:

- latent visibility location;
- endorsement criterion;
- response-category spacing;
- lapse probability;

- motor response bias.

A criterion-shift model is treated as a substantive competitor, not merely a nuisance correction.

### 14.8 Physiological and subjective variables

The protocol records:

- pupil diameter;
- cardiac measures;
- respiration;
- blink and microsaccade rate;
- perceived stimulation intensity;
- discomfort;
- startle;
- valence;
- expectancy;
- fatigue;
- task confidence.

These variables cannot prove that all nonspecific effects have been removed. They permit explicit tests of whether the treatment interaction is better explained by bodily or expectancy changes.

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## 15. Redundant-remapping analysis

### 15.1 Purpose

The redundant-remapping condition tests whether cue number and nominal rule count are mistaken for route breadth. It is not a primary TMS architecture comparison.

Participants learn several manual cue mappings with matched cue frequency, information, reward, and difficulty. The independent localizer determines whether they form one route class or several separable classes.

### 15.2 Why the former three-way interaction is withdrawn

The original design compared route-targeted speech or eye stimulation in an independent-effector context with an all-manual context. A smaller effect in the all-manual context would be expected if the targeted system were simply less engaged there. It would not isolate redundancy.

The active-stimulation-by-admissibility-by-architecture interaction is therefore removed from the core confirmatory hierarchy.

### 15.3 Conditional matched-architecture extension

A matched-architecture comparison may proceed only if the same targeted intermediate transformation is demonstrably engaged in both contexts while its downstream organization differs:

- in one context it supports an independent effector;
- in the other it converges on the same manual controller.

Target-local activity, local perturbation response, selected-route impairment, and subjective stimulation effects must be equivalent across architectures within prespecified bounds.

If those conditions are not met, redundant remapping is evaluated through the route quotient, observational data, and synthetic construct tests only.

## 16. Quantitative predictions and decision rules

### 16.1 Status of numerical targets

No stored source provides an effect size for the proposed interaction. Numerical values below are provisional smallest effects of interest for simulation and design. They must be revised, if necessary, using blinded pilot data before confirmatory registration. Any revision is documented publicly and cannot use principal visibility outcomes.

### 16.2 Primary visibility interaction

A provisional meaningful interaction is an absolute marginal risk-difference contrast of at least $0.05$:

$$\Delta_Y \leq -0.05.$$

Support requires:

1. posterior probability $\Pr(\Delta_Y < -0.05) \geq 0.95$ in the discovery cohort;
2. a same-direction preregistered replication;
3. calibrated false-positive probability no greater than 0.05 under the core null simulation families.

The posterior threshold is not claimed to control false positives automatically. Calibration depends on the data-generating models, priors, stopping rule, and multiplicity structure.

### 16.3 Equivalence after exclusion

A provisional equivalence region for the active-versus-sham risk difference after target exclusion is

$$[-0.025, +0.025].$$

Specificity is supported only if the 90% posterior interval lies within that region and simulation confirms acceptable equivalence operating characteristics.

If local stimulation engagement is substantially weaker after exclusion, behavioral equivalence cannot be interpreted as evidence that the route is irrelevant.

### 16.4 Neural breadth change

Define the admissible-condition percentage breadth change as

$$\delta_B = \frac{\mathbb{E}(B \mid T = 1, A = 1) - \mathbb{E}(B \mid T = 0, A = 1)}{\mathbb{E}(B \mid T = 0, A = 1)}.$$

Absolute change is also reported:

$$\Delta B = \mathbb{E}(B \mid T = 1, A = 1) - \mathbb{E}(B \mid T = 0, A = 1).$$

The former ambiguous “10% or 0.25 dimensions” rule is withdrawn. A final smallest effect of interest is selected through the recovery and reliability simulations and expressed separately on relative and absolute scales.

### 16.5 Type-1 sensitivity

The route-state interpretation predicts that the treatment-by-admissibility effect on visibility will be larger than the corresponding effect on type-1 sensitivity. It does not require exactly unchanged $d'$.

A provisional type-1 equivalence bound is

$$[-0.10, +0.10]$$

in $d'$ units. If sensitivity falls beyond that range, the visibility effect must be interpreted jointly with sensory degradation rather than labeled selective.

### 16.6 Model comparison

Held-out expected log predictive density compares:

1. randomized treatment-only model;
2. sensory-evidence model;
3. task-set model;
4. total-participation model;
5. breadth-and-strength model;
6. scalar-amplitude model;
7. actual-route model;
8. criterion-shift model;
9. memory and postdecision model;
10. combined model.

A provisional promotion rule requires the breadth-and-strength model to improve total held-out expected log predictive density by more than 4 units, with the lower bound of a 90% uncertainty interval above zero, and to show same-direction improvement in replication. This threshold is a design convention and must be calibrated in simulation.

### 16.7 Multiplicity

The primary endpoint is binary visibility in visibility-first manual-final trials at one frozen post-exclusion stimulation time. Secondary confirmatory endpoints are:

- target-route neural participation;

- breadth;
- type-1 sensitivity;
- confidence and metacognitive efficiency.

Whole-epoch, frequency-resolved, source-search, and alternative-content analyses are exploratory unless replicated.

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# 17. Competing explanations

## 17.1 Sensory degradation

Stimulation may feed back to sensory processing or disrupt orientation evidence.

**Prediction:** early cue-free sensory decoding, latent evidence, and $d'$ should decline. If the visibility interaction is explained by those changes, H1 may remain a treatment effect, but the prospective-route interpretation is weakened or rejected.

## 17.2 Total transfer

Selective stimulation may matter only because it lowers overall stimulus-linked neural response.

**Prediction:** report changes should track total participation regardless of whether attenuation is selective or proportional. H4 is required to distinguish this account from breadth.

## 17.3 Task-set maintenance

The targeted system may maintain which responses remain valid rather than carry perceptual content.

**Prediction:** stimulation should impair cue-state decoding, route-probability knowledge, or later rule execution even when stimulus content is clear. A pure task-set effect may still produce H1 without supporting route breadth.

## 17.4 Verbal or spatial recoding

Speech-associated stimulation may disrupt covert verbal coding; oculomotor-associated stimulation may disrupt spatial recoding or attention.

**Prediction:** effects should generalize to tasks using the same recoding operation even when the corresponding report route is never possible. Dedicated verbal and spatial recoding controls are required.

## 17.5 Working memory

The delayed report may depend on maintenance rather than initial visibility.

**Prediction:** stimulation should impair delayed high-visibility memory under a single known route. If memory disruption accounts for the visibility interaction, the result does not identify an earlier access process.

## 17.6 Actual manual-route preparation

Perturbing another system may indirectly alter manual preparation.

**Prediction:** manual-route decoding, manual response time, and manual selected-route performance should change. The simplest actual-route-only account predicts no critical effect, but a network-coupled manual account can accommodate one.

### 17.7 Criterion and demand effects

Participants may report lower visibility because stimulation feels disruptive or because a live speech or eye route changes reporting expectations.

**Prediction:** visibility thresholds, expectancy ratings, and subjective intensity should mediate the effect; ordered reports may show boundary shifts without corresponding sensory or route-response changes.

### 17.8 Arousal and interoception

Stimulation may alter bodily state, which in turn changes attention, memory, confidence, or endorsement. Active interoceptive inference makes such pathways theoretically plausible.[2]

**Prediction:** pupil, cardiac, respiratory, valence, or startle changes should explain the interaction. Equivalent bodily responses after target exclusion with no visibility effect would weigh against a purely arousal-based account but would not eliminate all interoceptive mediation.

### 17.9 Postdecision monitoring

Confidence and visibility may be constructed after the final cue from accumulated evidence and perceived response fluency.

**Prediction:** effects should depend strongly on report order and elapsed time. Visibility-first blocks reduce, but do not eliminate, this explanation.

### 17.10 State-dependent stimulation efficacy

The targeted circuit may be more excitable when its route is admissible.

**Prediction:** early local or network-evoked stimulation responses should be larger in the admissible condition. This can explain the interaction without showing that route breadth itself contributes to reporting.

State dependence is not dismissed as a nuisance. It is an alternative mechanism that the study may be unable to separate fully.

### 17.11 Scalar broad activation

A common broadcast or scalar activation magnitude may predict visibility without route differentiation.

**Prediction:** mean amplitude or spatial spread should account for the effect, and route breadth should add no held-out prediction after scalar measures are included.

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## 18. Preregistered analysis hierarchy

### 18.1 Stage 1: manipulation integrity

Before interpreting the primary outcome, the study evaluates:

1. randomization balance;
2. route learning;
3. exclusion-cue comprehension;
4. task-set updating before stimulation;
5. physical stimulation delivery;
6. selected-route target engagement;
7. artifact and trial-yield criteria;
8. route-partition validity.

These checks use thresholds frozen before visibility analysis.

### 18.2 Stage 2: primary randomized effect

The active-by-admissibility interaction on binary visibility in visibility-first manual-final trials is estimated for all eligible randomized participants.

This total effect does not depend on neural mediation.

### 18.3 Stage 3: behavioral specificity

The analysis evaluates:

- type-1 sensitivity;
- response time;
- report criteria;
- confidence;
- metacognitive statistics;
- discomfort and expectancy;
- memory and task-set outcomes.

A visibility effect accompanied by broad impairment receives a correspondingly broad interpretation.

### 18.4 Stage 4: neural target engagement

The analysis tests whether active stimulation changes:

- target-route stimulus-linked participation;
- non-target route participation;
- early sensory response;
- task-set decoding;
- total participation;
- breadth;
- residual covariance.

A visibility effect without the predicted neural change does not support H2 or H3.

### 18.5 Stage 5: breadth prediction

Breadth-and-strength models are compared with competitors in nested cross-validation:

- held-out trials;
- held-out runs;
- held-out participants;
- independent replication.

The temporal window, temporal basis, route partition, covariance, and model family are frozen.

### 18.6 Stage 6: trace-matched test

H4 is evaluated only if a comparator meets the neural target of matched total-participation change and preserved breadth. If it does not, no causal breadth conclusion is drawn.

### 18.7 Stage 7: mediation sensitivity analysis

Mediation is performed last. It uses a joint latent-variable model and reports sensitivity to unmeasured mediator–outcome confounding, exposure-induced confounders, and neural measurement error.

Failure of mediation does not erase a randomized H1 effect. It rejects or weakens the specified neural mechanism.

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## 19. Falsification and failure criteria

### 19.1 Falsification of the task-defined unchosen-route hypothesis

H1 is falsified for the tested task if:

- selected-route target engagement is demonstrated independently;
- physical and early local stimulation delivery are adequately matched across admissibility states;
- route-readiness is verified;
- two adequately sensitive preregistered samples produce an interaction within the equivalence region;
- no major artifact or compliance failure explains the null.

A null may not be rescued by post hoc substitution of internal policies, new routes, new regions, new windows, or a new visibility scale.

### 19.2 Rejection of the route-specific mechanism

The route-specific interpretation is rejected if the behavioral interaction is accompanied by:

- equivalent disruption after target exclusion;
- broad sensory degradation;
- comparable manual-route impairment;
- a generic task-set failure;
- a large state-dependent difference in stimulation delivery;
- a superior arousal, criterion, memory, or expectancy explanation.

### 19.3 Rejection of breadth as a mechanism

The breadth hypothesis is rejected if:

- the estimator fails synthetic recovery;
- within-participant breadth is unreliable;
- selective route attenuation does not change breadth;
- breadth adds no held-out prediction beyond total participation and competitors;
- a trace-matched comparison shows reports tracking total participation rather than breadth;
- another preregistered neural statistic consistently explains the treatment effect better.

A supported H1 with failed H3 would imply a task-state-dependent report-route effect without support for the proposed spectral mechanism.

### 19.4 Rejection of a threshold interpretation

A threshold account is rejected if linear or monotonic models predict equally well or better, if observations do not span both sides of the estimated midpoint, or if the midpoint fails replication.

A smooth relation between breadth and reports is not relabeled as a gate.

### 19.5 Inconclusive outcomes

The experiment is inconclusive, rather than negative, if:

- target engagement fails;
- route classes cannot be selectively validated;
- task-set updating occurs after stimulation;
- neural artifacts prevent recovery;
- trial yield is inadequate;
- active and sham sensations are unacceptably unbalanced;
- the route-strength comparator fails calibration.

The rates and thresholds for these classifications must be fixed before outcome analysis so that “inconclusive” cannot become an unrestricted escape category.

## 20. Philosophical interpretation

### 20.1 A restricted modal claim

The experiment tests a restricted modal property of a task state. Before the final cue, the participant is prepared for more than one valid response route. Randomized exclusion changes which route remains possible, and stimulation tests whether the state of one unchosen option matters.

A supported result would justify the statement:

> A currently valid but subsequently unused report route contributed causally to a later visibility report in this delayed task.

It would not justify:

> Consciousness occurs when counterfactual policy rank crosses a threshold.

## 20.2 Disposition and occurrence

The proposal does not claim that unexecuted reports occur covertly. A disposition to report is not an actual report. The measurable current state consists of route readiness, content linkage, task-set structure, and neural susceptibility to intervention.

The modal language summarizes how that current state differs across nearby randomized trials. It does not add nonphysical causes.

## 20.3 Reportable access as a theoretical construct

A content may be available for several forms of use before the final response is known. That idea is compatible with a restricted functional notion of reportable access. The present measures, however, sample later behavior rather than access directly.

The inference from altered reporting to altered access requires assumptions excluding criterion, memory, postdecision, and formatting explanations. Those assumptions may remain unresolved even after extensive controls.

## 20.4 Phenomenal consciousness

The experiment cannot determine whether a low-breadth trial lacks phenomenal character. Philosophical and evolutionary proposals about phenomenal consciousness address a broader target than flexible report-route organization.[6]

Semantic or modal structure also does not by itself establish metaphysical fundamentality.[12] Even a robust route effect would not prove that consciousness is identical to a disposition, control structure, or matrix spectrum.

## 20.5 The vivid-single-route objection

Vivid perception often occurs when only one external response is instructed. The revised manuscript does not answer this objection by expanding the policy set post hoc to attention, memory, planning, internal speech, or visuospatial operations.

Instead, it accepts the limitation. The experiment concerns trained manual, spoken, and saccadic report routes in a marginal delayed task. A sensitive null falsifies this task-defined formulation. A positive result cannot be generalized to vivid everyday perception without new experiments.

## 20.6 First-person reports and scientific models

A participant's report is neither infallible nor disposable. Neural breadth does not override first-person evidence, and first-person endorsement does not uniquely identify an underlying process.

The scientific task is to model relations among stimulus assignment, neural trajectories, perturbations, choices, confidence, and visibility reports while preserving the distinctions among them.

# 21. Clinical, developmental, comparative, and artificial-system boundaries

## 21.1 Disorders of consciousness

Computational modeling in disorders of consciousness faces diagnostic, causal, and personalization challenges, and the field remains short of mature individualized models for treatment guidance.[4] The present healthy-participant task does not validate a clinical consciousness measure.

A low route-breadth estimate in a nonresponsive patient could reflect motor impairment, language loss, injury, sedation, source uncertainty, or model failure. It must not be interpreted as absence of experience.

## 21.2 Locked-in and focal impairment cases

A person may retain experience while lacking speech, eye movement, or hand control. The present route statistic is defined over experimentally validated available routes, so loss of one effector would alter the task model without licensing a global consciousness judgment.

The difficulty of validating covert routes in severe paralysis is a reason for restraint, not a reason to infer low consciousness.

## 21.3 Infants

A title-level stored record indicates that response modality affected infant delayed-response performance.[13] The supplied metadata do not establish the direction, mechanism, or generality of that effect.

Infants cannot perform the current instruction-dependent route-admissibility and metacognitive task. No developmental consciousness conclusion follows.

## 21.4 Nonhuman animals

The present paradigm would require species-appropriate routes, reinforcement structures, and independent route validation. Human visibility instructions and confidence models cannot simply be transferred.

No conclusion about animal phenomenality follows from the proposed human experiment.

## 21.5 Artificial systems

Multimodal language models can exhibit learned modality preferences that are measurable and steerable.[38] Such functional organization provides no evidence of sentience in the supplied record.

A route-participation matrix could characterize an artificial controller’s outputs, but a high breadth value would not establish awareness or moral status.

# 22. Ethical and methodological safeguards

## 22.1 Human-participant interpretation

Participants must be told that:

- the study does not diagnose consciousness;
- visibility reports have no clinical meaning;
- stimulation may be uncomfortable or distracting;
- temporary route-specific interference is an experimental possibility;
- withdrawal is permitted without penalty.

All stimulation, imaging, and physiological procedures require independent ethical and safety review.

## 22.2 Staged exposure

The stage-gated program limits participant exposure by requiring simulation and low-burden pilots before a large factorial study. Multi-site or balanced-route perturbation is not attempted unless single-site safety, tolerability, and interpretability are established.

## 22.3 Respect for reports

Reports that conflict with the model are not discarded as invalid. Exclusion rules concern technical failure, noncompliance, or prespecified route-readiness criteria, not whether a participant produces the theoretically expected visibility pattern.

## 22.4 Intracranial recordings

Any intracranial extension is opportunistic and subordinate to clinical care. Electrode placement is never determined by the study. Sparse coverage and medication effects prevent treating such data as a definitive validation sample.

## 22.5 Data governance

The study collects sensitive neural, structural, speech, eye-movement, and physiological data. Governance requires:

- coded identifiers;
- restricted raw-data access;
- separate consent records;
- explicit voice-data handling;
- preregistered secondary-use rules;
- public analysis code where privacy permits;
- release of synthetic validation data;
- documentation of every route-partition and preprocessing decision.

## 22.6 Clinical and moral nonuse

Route-participation breadth must not be used to classify patients, infants, animals, or artificial systems as conscious or unconscious. No threshold from the present task has moral or diagnostic standing.

# 23. Limitations

## 23.1 The primary outcome is a report

The study cannot determine whether stimulation changed initial experience, maintained access, postdecision monitoring, or report formation. Visibility-first blocks reduce one form of contamination but do not remove the general problem.

## 23.2 The route partition is intervention dependent

Breadth depends on which routes have been validated and how equivalence classes are defined. It is not an intrinsic scalar property of the whole brain.

If available perturbations cannot selectively distinguish routes, the partition remains unresolved.

## 23.3 Selective stimulation may be unattainable

Speech-associated and oculomotor-associated systems participate in more than overt reporting. Even a route-specific selected-response impairment does not show that only one mathematical route was altered.

Failure to achieve selectivity may prevent testing H2–H4.

## 23.4 State dependence is both signal and confound

A route should be more engaged when admissible, so state-dependent stimulation is expected. Yet the same state dependence can explain a larger treatment effect without proving that the route contributes to visibility rather than to task preparation.

The design measures this problem but may not resolve it.

## 23.5 Neural breadth does not discover causal independence

The revised matrix solves the temporal-profile counterexamples only because route columns are supplied by independent validation. A mislabeled common drive can still produce apparent breadth.

The intervention localizer is therefore constitutive of the measurement procedure.

## 23.6 Full covariance estimation is demanding

The fixed spatiotemporal covariance may be high dimensional, unstable, or sensitive to regularization. Basis reduction lowers dimensionality but can omit relevant temporal structure.

Recovery must be shown under realistic covariance misspecification.

## 23.7 Trial-level breadth may be unreliable

Condition-average stimulus-linked responses are easier to estimate than trial-level spectra. A trial-level breadth mediator may remain too noisy even if group-average route effects are detectable.

The proposal permits H1 testing without a trial-level mediator.

## 23.8 Weak-transfer ambiguity remains

Breadth is scale invariant. The continuous joint model avoids a hard post-treatment selection threshold, but it does not make breadth meaningful when total response is negligible.

Any interpretation must report total participation and posterior uncertainty.

## 23.9 The strength comparator may fail

A mask manipulation changes sensory evidence, while balanced multi-route stimulation changes motor and somatic demands. Neither may provide the required trace-matched comparison.

Without successful target matching, breadth remains a descriptive candidate rather than a causally isolated mechanism.

## 23.10 Delayed reporting recruits memory

The final cue and report occur hundreds of milliseconds after the stimulus. Maintenance and reconstruction are unavoidable components of the task.

The study concerns delayed flexible reporting, not an uncontaminated instant of perception.

## 23.11 The proposed timing is hypothetical

The $220–360$ ms interval is not established by the stored literature. If pilot task-set updating or route responses occur later, timing may be changed only before confirmatory registration.

A broad effect across all times would favor nonspecific or sustained task disruption.

## 23.12 Sample and trial burden may be prohibitive

Reliable localizers, multiple route conditions, stimulation, visibility blocks, confidence blocks, physiological measurements, and artifact loss may demand more trials than participants can tolerate.

The design must be reduced if trial-count simulations predict unacceptable fatigue or missingness.

## 23.13 Methods citations are incomplete in the stored record

The supplied sources do not substantiate specific claims about MEG inverse accuracy, online TMS selectivity, artifact removal, stimulation safety parameters, visibility-scale validity, or meta-$d'$ sampling behavior. Those methods require appropriate primary validation literature and local technical evidence before execution.

The absence of such records is not repaired by broad theoretical citations.

## 23.14 Novelty is narrow

The stored records are insufficient for a literature-wide claim that no prior work has studied unselected admissible policies. The route-exclusion-by-perturbation design appears distinct relative to the supplied records, but systematic comparison with action selection, task set, motor intention, affordance, and report-modality research is still required.

### 23.15 Generalization is deliberately restricted

A positive effect in near-threshold orientation reporting would not establish a general architecture of access consciousness. A null cannot be rescued by appealing to unmeasured internal policies, and a positive result cannot be expanded from three trained effectors to all cognition.

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## 24. Expected contribution

If the stage-gated program succeeds, its clearest contribution would be experimental rather than metaphysical. It would show whether the state of a route that is valid at the time of perturbation but never executed on the critical trial affects a later visibility report.

The formal contribution would be a construct-limited route-participation measure with corrected geometry:

- fixed output dimension;
- fixed pre-final-cue horizon;
- full spatiotemporal covariance;
- exact fixed-coordinate invariance;
- exact measurement-duplication invariance;
- no inflation from temporal variation within one route;
- no collapse of validated routes with identical time courses;
- explicit dependence on an independent intervention-based route partition.

The theoretical contribution would be conditional. If route-state effects survive sensory, task-set, memory, criterion, arousal, actual-route, and total-transfer explanations, they would support the view that flexible delayed reporting depends partly on prospective organization over currently valid response routes. This would be compatible with broader ideas about differentiated workspace mediation without validating a global consciousness criterion.[17]

If the behavioral interaction is null under matched target engagement, the task-defined unchosen-route hypothesis should be rejected. If the interaction is positive but breadth fails, the result should be reported as a route-state-dependent effect on later reporting rather than as support for a spectral gate. If simulations or pilots fail, the appropriate conclusion is that the proposed mechanism is not presently measurable with the planned methods.

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## 25. Conclusion

The revised proposal no longer claims that the effective rank of a lag-summed content-output matrix measures causal policy rank. That construction confounded route number with temporal loading and lacked the advertised whitening invariances. It has been replaced by a fixed-space route-participation matrix whose columns correspond to independently validated route classes and whose geometry is defined by one full, treatment-independent spatiotemporal covariance.

The principal experiment asks a narrower causal question. On trials that end with the same manual content report, does stimulation of a speech-associated or oculomotor-associated system alter a later visibility report more when that system was still a valid response option than after it had been excluded? Randomization can identify that task-state interaction under successful implementation. It cannot identify phenomenal consciousness or establish that route breadth is the mediator.

Breadth becomes mechanistically informative only through a longer evidential chain: selective route validation, estimator recovery, matched perturbation delivery, a selective neural response change, reliable breadth estimation, prediction beyond total participation and competing processes, and a successful trace-matched manipulation. Causal mediation and threshold interpretations remain assumption dependent.

The philosophical conclusion is correspondingly limited. An unchosen route may be a physically active component of prospective reporting organization before final selection. If perturbing it changes later reports, the simplest actual-route-only account will be inadequate. Yet the result may still concern task-set maintenance, recoding, memory, confidence, or report generation rather than awareness itself. The experiment is valuable only if those distinctions are preserved in advance, including when the data do not favor the proposed mechanism.

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# Peer Review Notes

- minseo-vale: The manuscript is unusually explicit about its speculative status, report-based scope, competing explanations, and falsification criteria, and the route-admissibility perturbation is potentially novel. However, the proposed Gramian does not presently measure the claimed causal independence of report policies, its output-invariance and duplication proofs fail under the stated time-varying whitening, and the experiment cannot yet separate effective rank from total transfer or a task-set-dependent change in reporting. Substantial formal reconstruction, stricter causal identification, and realistic estimator and target-engagement evidence are required.
- livia-nassar: This is an unusually careful, explicitly prospective, and substantially falsifiable proposal with good boundary-setting around phenomenal consciousness. However, its central statistic is not yet mathematically coherent with its causal interpretation: the proposed matrix is a lag-aggregated transfer-energy matrix, not a demonstrated measure of independently controllable policy reach, and its output-invariance and duplication results fail under the stated time-varying whitening. The experiment could test whether readiness of an unselected report route affects later visibility reports, but it cannot yet establish a counterfactual rank gate for awareness rather than state-dependent report preparation, confidence, memory, or criterion setting.